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1 INTRODUCTION

1.1 STUDY OBJECTIVES

- To define, describe, and forecast the global microbiome sequencing services market based on service, type, technology, end user, and region
- To provide detailed information regarding the major factors influencing the market growth (such as drivers, restraints, challenges, and opportunities)
- To strategically analyze micromarkets¹ with respect to individual growth trends, prospects, and contributions to the overall microbiome sequencing services market
- To analyze the opportunities in the market for stakeholders and provide details of the competitive landscape for market leaders operating in the microbiome sequencing services market
- To forecast the revenue of the market with respect to five regional segments—North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa
- To profile the key market players and comprehensively analyze their service portfolios, market positions, and core competencies²
- To track and analyze competitive developments such as services launches, expansions, agreements, collaborations, partnerships, and acquisitions in the microbiome sequencing services market

1.2 MARKET DEFINITION

Microbiome sequencing is applied to decode the genetic material of all the microorganisms in a given sample which may contain bacteria, viruses, or fungi. Whole-genome sequencing (WGS) and amplicon sequencing are the two key microbiome sequencing services. Sequencing-based detection of microorganisms exhibits different advantages over traditional approaches, some of which are mentioned below:

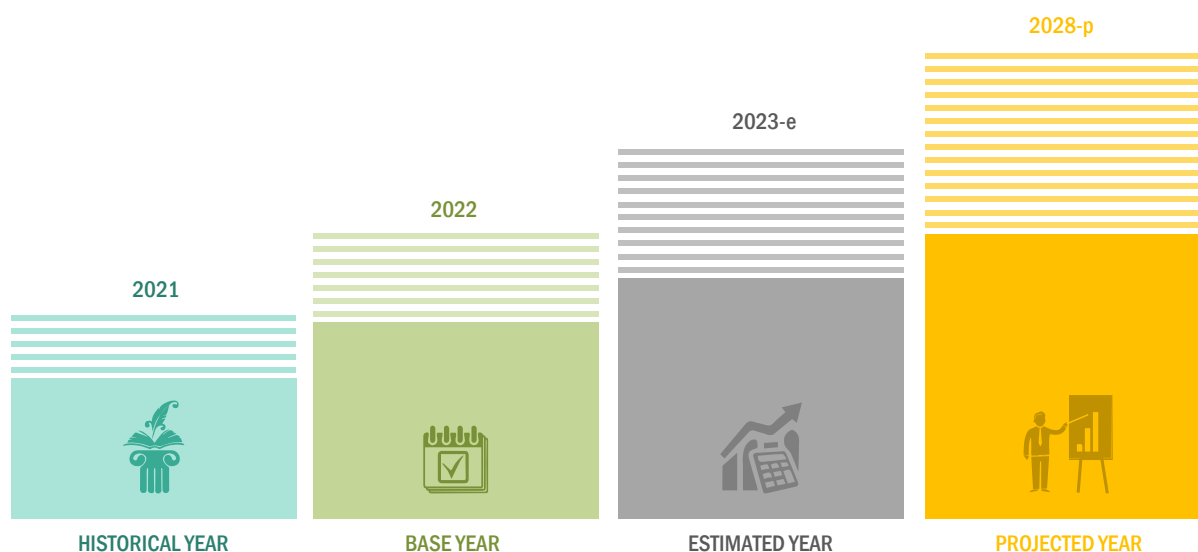
- Detection of anaerobic bacteria
- More cost-effective microbiome characterization of multiplex microbial samples
- Identification of non-cultivable bacteria
- Estimate the relative abundance of microbes in a sample
- Profile hundreds of microorganisms in a single analysis
- Examine complex microbiomes and microbiota communities

The microbiome sequencing services market comprises services employed for sample preparation, sequencing, library preparation, and data analysis suitable for different samples, including human microbiome samples and environmental samples.

1. Micromarkets are the further segments and subsegments of the microbiome sequencing services market.

2. Core competencies of companies are captured in terms of the key developments, market shares, and key strategies adopted to sustain their positions in the market.

1.3.2 YEARS CONSIDERED



Note: For market estimates, the calendar year has been considered, whereas, for company profiles, the fiscal year has been considered.

1.4 CURRENCY

- The currency used in this report is the United States Dollar (USD), with market sizes indicated in terms of USD million/billion.
- For companies reporting their revenues in USD, the revenues were taken from their respective annual reports/SEC filings.
- For companies reporting their revenues in other currencies, the average annual currency conversion rate was used for that particular year to convert the value to USD.

1.5 LIMITATIONS

- The market study does not cover the market size in terms of volume, as it is a service-oriented market.
- A limited number of industry experts participated in primary research from Africa, the Middle East, and Latin America due to specific language barriers. In such cases, the geographic market size was derived based on weightages assigned to these markets based on the qualitative insights from global industry experts, the analysis of demographic parameters, and the adoption trends for microbiome sequencing observed in these markets.
- Financial information for privately held and unlisted companies was not available in the public domain. Hence, their financial details are not provided in the company profiles section.
- Company developments not reported in the public domain are not included in this report.
- Market shares are estimated on a year-on-year basis (2020 to 2022) using historical data of companies and secondary sources.
- Several strategic approaches and recommendations from industry experts were utilized to collate the research study to overcome the limitations and maintain the effectiveness of the study.

1.6 STAKEHOLDERS

- Research Institutes
- Pharmaceutical and Biotechnology Companies
- Microbiome Sequencing Service Providers
- Metagenomic Sequencing Service Providers
- NGS Service Providers
- Venture Capitalists and Investors
- Contract Research Organizations
- Government Associations
- Healthcare Associations/Institutes
- Business Research and Consulting Service Providers

1.7 SUMMARY OF CHANGES

Refinements in the segments of the global microbiome sequencing services market:

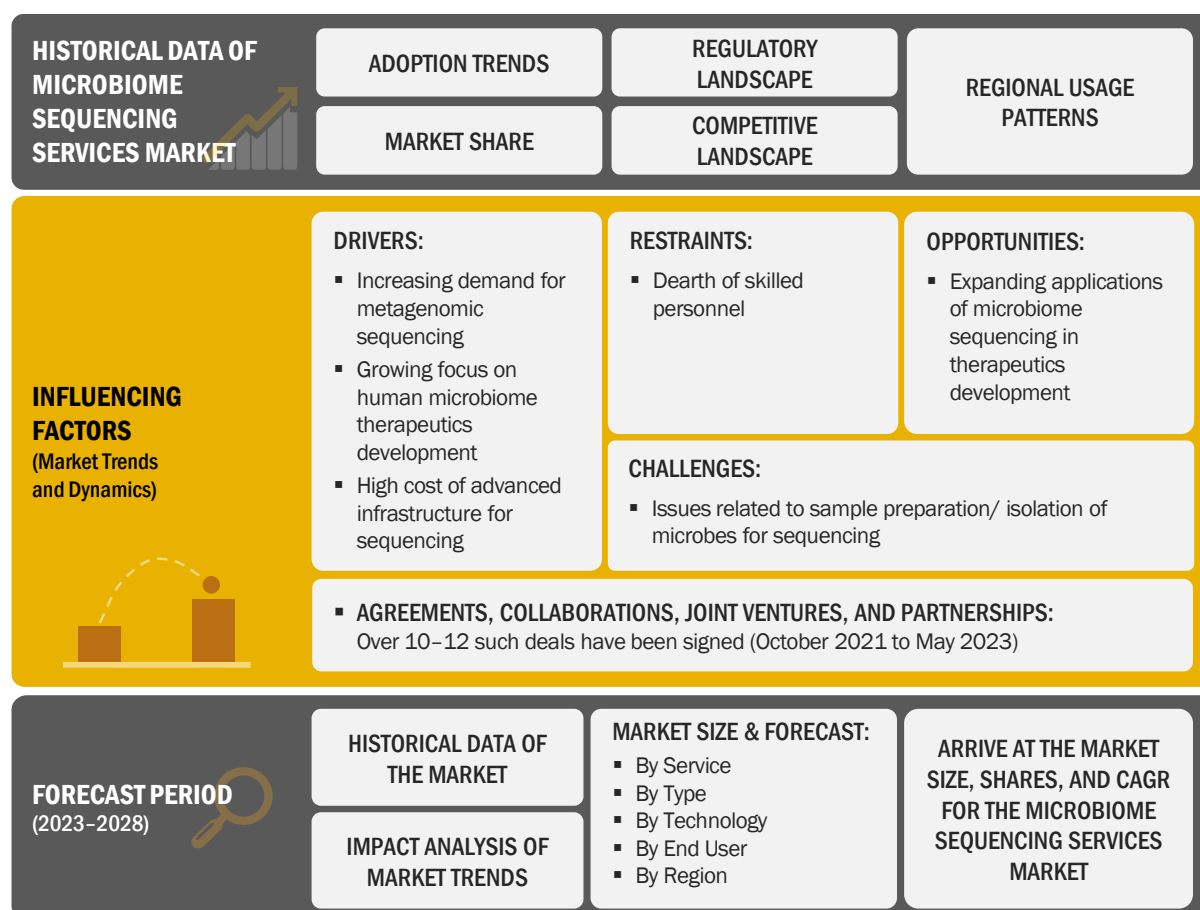
- Market sizes are updated for the base year 2022 and forecasted from 2023 to 2028.
- The new edition of the report provides updated financial information till 2022. This will help in the easy analysis of the present status of profiled companies in terms of their financial strength, profitability, key revenue-generating region/country, and business segment focus in terms of the highest revenue-generating segment.
- Recent developments are helpful in understanding market trends and the growth strategies adopted by players in the market. The number of service launches has increased in the last three years (October 2020 to April 2023).
- Tracking the service portfolios of prominent market players helps analyze the major services in the microbiome sequencing services market. The new edition of the report provides an updated service portfolio of the companies profiled in the report.
- Key market strategies, market share analysis, and competitive leadership mapping have been updated in the competitive landscape section of the report.
- The updated version of the report includes the revenue share analysis of key market players from 2020 to 2022.
- The current edition of the report includes an assessment of the economic and financial recession and geographical conditions such as supply chain disruptions, the Russia-Ukraine war, along with the effects of inflation on the services market. The impact of the recession on year-on-year growth during the forecast period (2023 to 2028) has also been considered.

2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

This market research study involved the extensive use of secondary sources, directories, and databases to identify and collect information useful for this technical, market-oriented, and financial study of the global microbiome sequencing services market. In-depth interviews were conducted with various primary respondents, including key industry participants, subject-matter experts (SMEs), C-level executives of key market players, and industry consultants, to obtain and verify critical qualitative and quantitative information and assess market growth prospects. The global size of the microbiome sequencing services market (estimated through various research approaches) was then triangulated with inputs from primary research to arrive at the final market size.

FIGURE 1 RESEARCH DESIGN

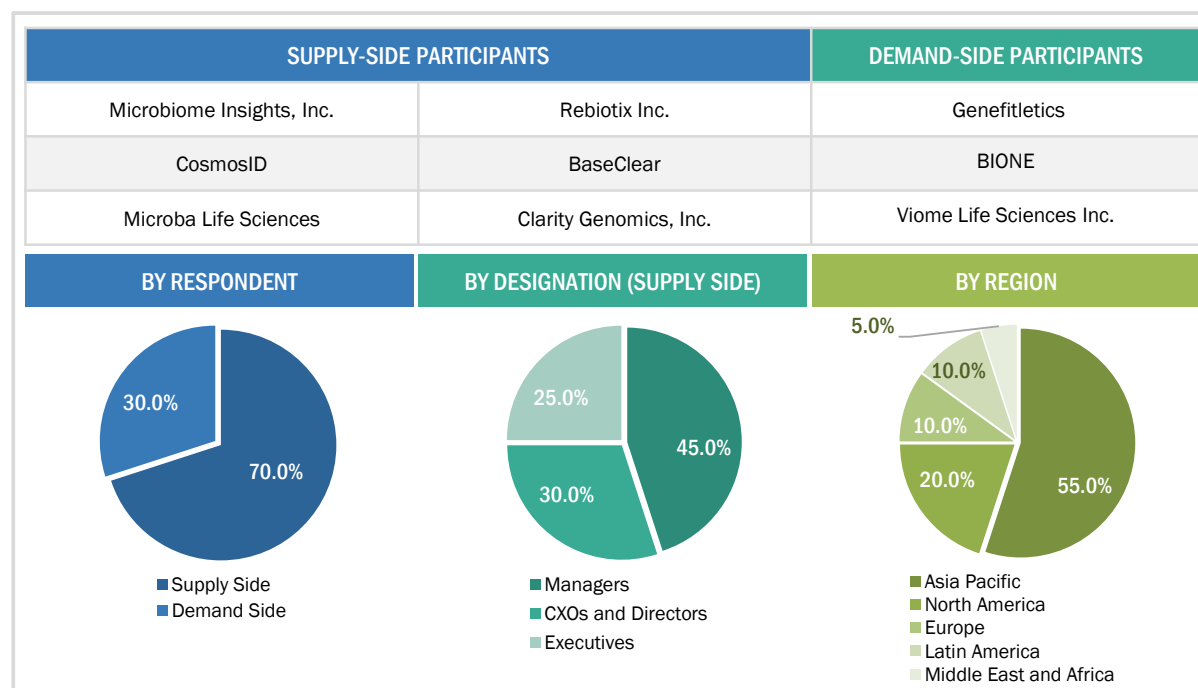


Sources: Secondary Research and MarketsandMarkets Analysis

2.1.2 PRIMARY DATA

Extensive primary research was conducted after acquiring basic knowledge about the global microbiome sequencing services market through secondary research. Several primary interviews were conducted with market experts from both the demand side (personnel from pharmaceutical and biopharmaceutical companies, medical device companies, and academic institutes) and supply side (C-level and D-level executives, product managers, and marketing and sales managers of key manufacturers, distributors, and channel partners, among others, from tier 1 and tier 2 companies engaged in offering services) across five major regions—North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Approximately 70% and 30% of primary interviews were conducted with supply-side and demand-side participants, respectively. This primary data was collected through questionnaires, e-mails, online surveys, personal interviews, and telephonic interviews.

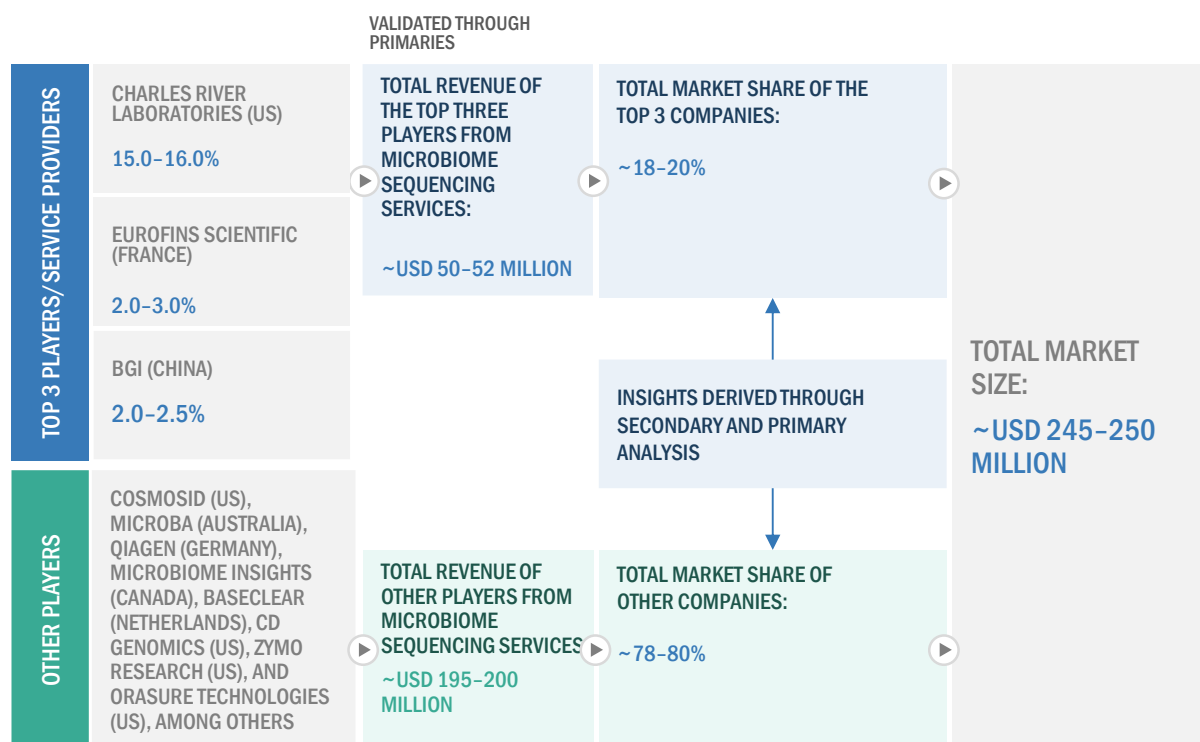
FIGURE 2 MICROBIOME SEQUENCING SERVICES MARKET: BREAKDOWN OF PRIMARIES



Primary research was used in this report to:

- Validate the market segmentation defined through a service portfolio assessment of the leading players in the microbiome sequencing services market
- Understand key industry trends and issues defining the strategic growth objectives of market players
- Gather both demand and supply-side validation of the key factors affecting the market growth (such as market drivers, restraints, challenges, and opportunities)
- Validate assumptions for the market sizing and forecasting model used for this study
- Understand the market position of leading players in the microbiome sequencing services market and their market shares/ranking

FIGURE 4 MARKET SIZE ESTIMATION: APPROACH 1 (COMPANY REVENUE ANALYSIS-BASED ESTIMATION), 2022



Source: MarketsandMarkets Analysis and Primary Insights

For the calculation of the global market value, the segmental revenue was arrived at based on the revenue mapping of major players active in the microbiome sequencing services market. This process involved the following steps:

- A list of the major global players operating in the microbiome sequencing services market was generated.
- The annual revenues generated by major global players from the microbiome sequencing services market, or the nearest reported business unit/category, were mapped.
- The revenues of the top 3 players in the microbiome sequencing services market were identified.
- The revenue of the top 3 players was assumed to be approximately 18-20% in 2022 and added up to 100% to derive the global value of the microbiome sequencing services market.

2.5 RESEARCH ASSUMPTIONS

- Market values have been rounded off to the nearest decimal. Due to this, there may be slight differences between actual totals and the totals mentioned in market tables.
- Region-wise inflation/deflation for service pricing during 2023–2028 has not been considered.
- It has been assumed that dollar fluctuations will not affect the growth forecast to a significant extent.
- Revenues of private companies acquired from paid databases were assumed to be in line with actual company financials.
- A positive economic climate will continue in emerging countries until 2028, and the recession of 2023 will have a low impact on the microbiome sequencing services market.
- There will be no substantial changes in government regulations that will significantly impact the market.
- All market segments and subsegments listed in this report are considered to be mutually exclusive.
- Wherever the exact segmental market splits were not available, historical trends and inputs from primary respondents have been considered.

2.6 RISK ANALYSIS

DESCRIPTION	RISK ANALYSIS
The growth rate for each segment is considered based on recent market trends, key developments of companies, analysis of historical growth rates, and primary insights.	Moderate
Market sizes and shares for 2022 were estimated based on the financials available for nine months in company annual/quarterly reports and presentations for a few companies.	Low to Moderate

2.7 IMPACT OF RECESSION ON MICROBIOME SEQUENCING SERVICES MARKET

A recession is a period of temporary economic decline during which trade and industrial activities are reduced. It is generally identified by a decline in the GDP in two successive quarters. According to the World Economic Forum Forecast Report (October 2022), global GDP growth would slow from 6% in 2021 to 3.2% in 2022 and 2.7% in 2023. Alternatively, the same report suggests that global inflation would rise from 4.7% in 2021 to 8.8% in 2022 but decline to 6.5% in 2023 and to 4.1% by 2024.

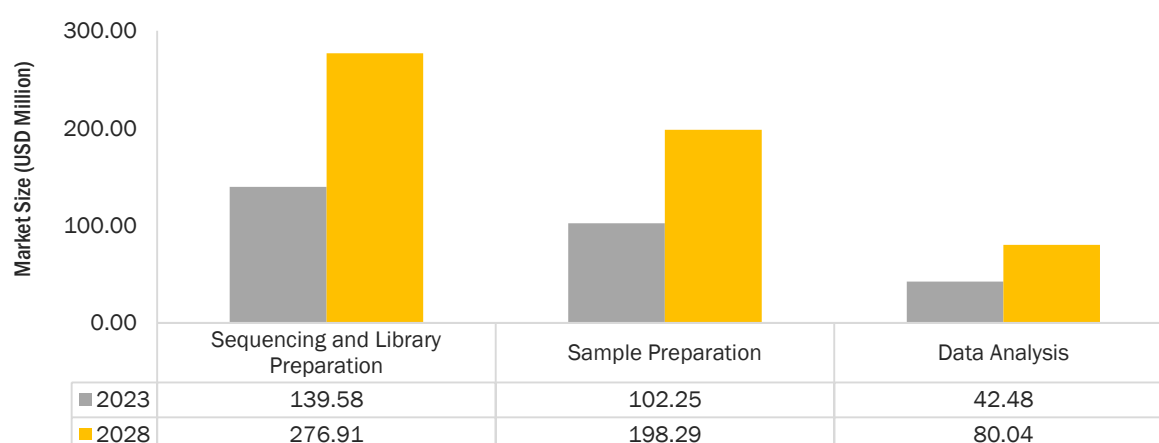
The key reasons behind this slowdown are the Russia-Ukraine war (which led to a severe energy crisis in Europe, where gas prices have escalated four-fold since 2021) and frequent lockdowns in China under its 'Zero COVID policy,' which impacted global trade and led to a sharp appreciation of the US dollar.

Based on analysis by the pharmaceutical and biotechnology industry and inputs from primary experts, the pharmaceutical and biotechnology industry is expected to withstand this downturn. The demand for healthcare products and services is expected to endure, and the recession will have a low to minimal impact on overall healthcare-oriented businesses.

3 EXECUTIVE SUMMARY

The microbiome sequencing services market is projected to grow from USD 284.31 million in 2023 to USD 555.24 million by 2028, at a CAGR of 14.3%. Growth in this market is majorly driven by the increasing demand for metagenomic sequencing, the growing focus on human microbiome therapeutics development, and the high cost of advanced infrastructure for sequencing. However, a dearth of skilled personnel is expected to hinder the growth of this market during the forecast period.

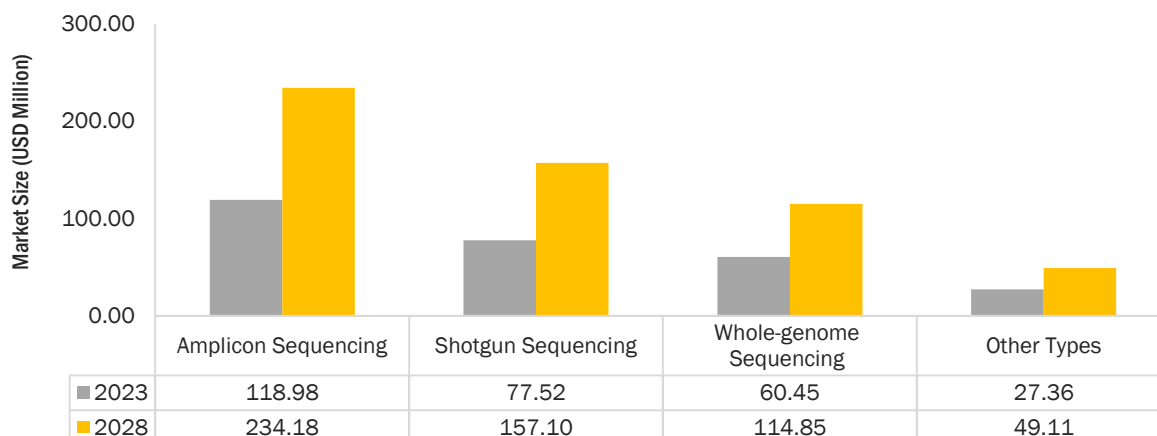
FIGURE 10 MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on service, the microbiome sequencing services market is segmented into sample preparation, sequencing and library preparation, and data analysis services. A majority of the service providers/key market players offer all three types of services as part of the microbiome analysis workflow. Variations in sample size, associated pricing, use of different technologies, and the strength of local players are the key factors driving the revenue generation capacity of the above-mentioned segments. In 2022, the sequencing and library preparation segment accounted for the largest market share. This segment is also estimated to grow at the highest CAGR during the forecast period.

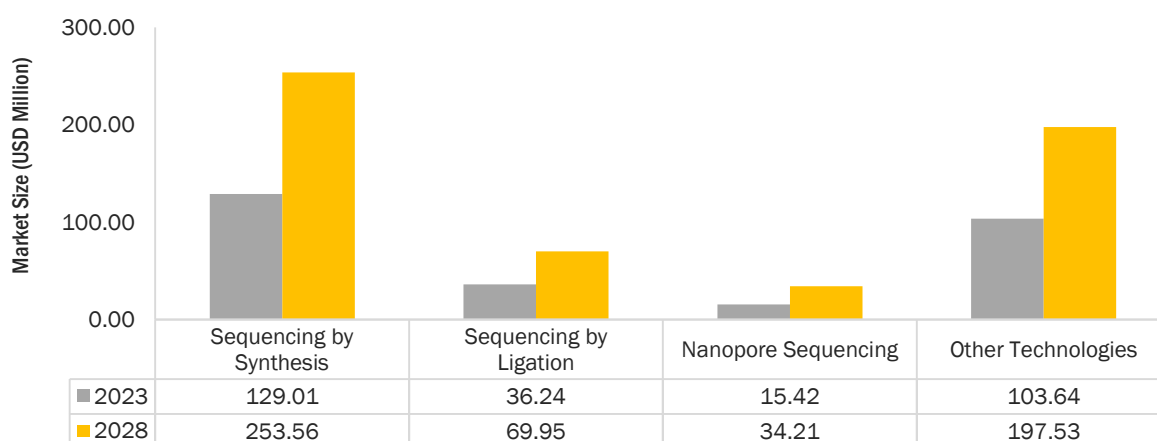
FIGURE 11 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on type, the global microbiome sequencing services market is segmented into amplicon sequencing, whole-genome sequencing, shotgun sequencing, and other types (including de novo sequencing, metagenomic sequencing, and metatranscriptomic sequencing). The amplicon sequencing segment accounted for the largest market share in 2022. The large share of this segment can be attributed to the high adoption of 16S rRNA microbiome sequencing and targeted sequencing. Key market players/service providers have built an innovative portfolio around these services.

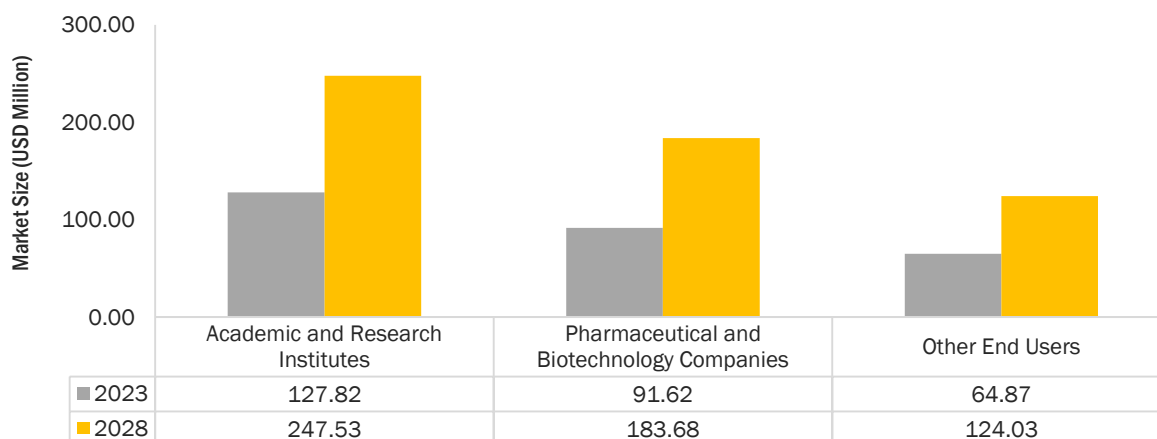
FIGURE 12 MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

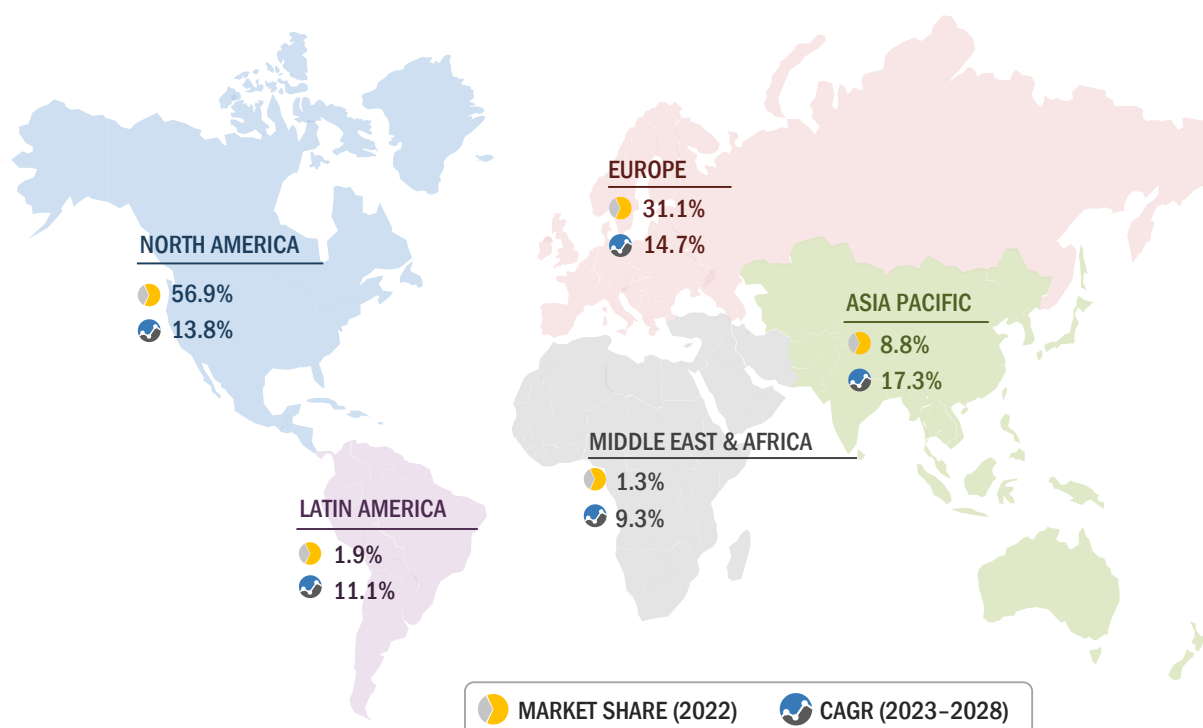
Based on technology, the global microbiome sequencing services market is segmented into sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and other technologies (semiconductor sequencing and Sanger sequencing, among other emerging technologies). The SBS segment accounted for the largest market share in 2022, while the nanopore sequencing segment is estimated to grow at the highest CAGR during the forecast period.

FIGURE 13 MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on end users, the global microbiome sequencing services market is segmented into pharmaceutical and biotechnology companies, academic and research institutes, and other end users (including small-scale CROs, CDMOs, hospitals, and clinics). The academic and research institutes segment accounted for the largest market share in 2022. This can be attributed to the high adoption of microbiome sequencing services and clinical research in these facilities.

FIGURE 14 GEOGRAPHICAL SNAPSHOT OF MICROBIOME SEQUENCING SERVICES MARKET

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

The microbiome sequencing services market is segmented into North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Factors such as established research infrastructure, technological advancements, supportive regulatory environment, robust biotechnology and healthcare industry, and the availability of funding and investment are driving the growth of the microbiome sequencing services market in North America. The microbiome sequencing services market in the Asia Pacific is expected to register the highest growth rate during the forecast period due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine.

The prominent players in the global microbiome sequencing services market are Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

5.7.4 BARGAINING POWER OF BUYERS

The bargaining power of buyers in the microbiome sequencing services market is moderate. Academic research institutes (one of the largest end users of sequencing products) are involved in bulk purchases of products and bulk sequencing of samples. This trend of bulk purchasing further increases the bargaining power of buyers in this market.

5.7.5 INTENSITY OF COMPETITIVE RIVALRY

The microbiome sequencing services market is fragmented in nature, with the top 3 players accounting for a share of ~20%. A large number of local players and small-scale CROs account for the remaining market share. These players operate in niche market segments. Industry players are expected to cash in on the various growth opportunities and gain maximum share in the market. This has created low-entry barriers for new players, and the existing players compete to expand their shares.

5.8 REGULATORY ANALYSIS

5.8.1 NORTH AMERICA

5.8.1.1 US

NGS sequencing technologies are being used predominantly in microbiome and metagenome sequencing. The increase in use is further supported by increasing advancements in genomics. Several federal agencies regulate NGS products, including the Food and Drug Administration (FDA), the Centers for Medicare and Medicaid Services (CMS), and the Federal Trade Commission (FTC).

CMS Regulations:

The CMS regulates clinical laboratories, including laboratories conducting clinical genetic testing, through its Clinical Laboratory Improvement Amendments of 1988 (CLIA) program. The CLIA established a certification that laboratories must pass to conduct clinical testing legally. The objective of the CLIA is to determine clinical testing quality, including verification of the procedures used and the qualifications of the technicians processing the tests.

FDA Regulations:

Until recent years, the FDA has practiced enforcement discretion for laboratory-developed tests (LDTs), which implies that LDTs are used in the clinic without the FDA's assessment of their analytical and clinical validity. However, due to rapid advances in NGS, the growing need for clinical genetic testing, the growth of direct-to-consumer (DTC) genomic testing, and concerns that unregulated tests pose a public health threat, the FDA is modifying its approach.

The FDA has developed new guidance to regulate NGS genetic tests and verify their analytical and clinical validity. The agency has also drafted guidance proposing a new regulatory framework for LDTs. However, FDA guidance only represents the FDA's current thinking on a topic and is not legally binding for the FDA or the parties it regulates. However, adhering to FDA guidance is beneficial because it can streamline the regulatory process. The draft guidance is listed below. Since they are in draft form, they are not currently being implemented.

List of FDA Draft Guidance on Genetic Tests:

- Framework for Regulatory Oversight of Laboratory Developed Tests (2014)
- FDA Notification and Medical Device Reporting for Laboratory Developed Tests (2014)
- Discussion Paper on Laboratory Developed Tests (2017)
- Next-generation Sequencing (NGS) Draft Guidance

7 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE

KEY FINDINGS

- The amplicon sequencing segment accounted for the largest share of 41.0% of the global microbiome sequencing services market in 2022. This segment is projected to reach USD 234.18 million by 2028 from USD 118.98 million in 2023, at a CAGR of 14.5%.
- The shotgun sequencing segment is expected to grow at the highest CAGR of 15.2% from 2023 to 2028. This segment is expected to reach a value of USD 157.10 million by 2028.
- Whole-genome sequencing is the third-largest segment in terms of revenue, with a market share of approximately 21.0% in 2022.

7.1 INTRODUCTION

Based on type, the global microbiome sequencing services market is segmented into amplicon sequencing, whole-genome sequencing, shotgun sequencing, and other types (including de novo sequencing, metagenomic sequencing, and metatranscriptomic sequencing). The amplicon sequencing segment accounted for the largest market share of 41.0% in 2022. The large share of this segment can be attributed to the high adoption of 16S rRNA microbiome sequencing and targeted sequencing. Key market players/service providers have built an innovative portfolio around these services, including sample preparation and data analysis solutions.

TABLE 29 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	91.80	104.54	118.98	135.54	154.55	176.84	203.06	234.18	14.5%
Whole-genome Sequencing	47.29	53.48	60.45	68.39	77.44	87.98	100.31	114.85	13.7%
Shotgun Sequencing	59.09	67.70	77.52	88.83	101.89	117.27	135.44	157.10	15.2%
Other Types	21.85	24.46	27.36	30.62	34.29	38.52	43.41	49.11	12.4%
Total	220.03	250.17	284.31	323.39	368.16	420.61	482.21	555.24	14.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.2 AMPLICON SEQUENCING

7.2.1 HIGH ACCURACY OF 16S AMPLICON SEQUENCING TO DRIVE ADOPTION

This segment is inclusive of 16S rRNA sequencing, ITS sequencing, and targeted sequencing services. Amplicon sequencing is a highly targeted approach that allows researchers to analyze genetic variation in specific regions of a genome. The ultra-deep sequencing of PCR products (amplicons) allows efficient variant identification and characterization. 16S rRNA sequencing is the most common technique utilized for metagenomics studies. In this technique, the 16S rRNA genetic markers present in a bacterial genome are sequenced to study bacterial phylogeny and taxonomy from complex microbiomes or environments that are difficult or impossible to analyze. The presence of 16S rRNA genetic markers as a multigene family or operons in almost all bacteria and the large size of the 16S rRNA gene (~1,500 bp) makes it the most preferred metagenomic sequencing technique among end users.

The demand for targeted sequencing is anticipated to increase in metagenomic research activities. The targeted amplicon sequencing method involves sequencing a phylogenetically informative marker to identify organisms in community samples. The marker must be present across all samples with expected organisms to enhance taxonomic profiling through the amplification of the primer. A number of different markers are commonly used and vary by the taxonomy of interest, but the most commonly used targeted amplicon approaches include the 16S rRNA gene for bacterial samples, the 18S for eukaryotes, and ITS for fungal samples.

8.1 INTRODUCTION

Based on technology, the global microbiome sequencing services market is segmented into sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and other technologies (semiconductor sequencing and Sanger sequencing, among other emerging technologies). The SBS segment accounted for the largest market share (~45%) in 2022, while the nanopore sequencing segment is estimated to grow at the highest CAGR (17.3%) during the forecast period.

TABLE 50 MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	99.59	113.38	129.01	146.93	167.48	191.59	219.92	253.56	14.5%
Sequencing by Ligation	28.18	31.96	36.24	41.13	46.71	53.24	60.89	69.95	14.1%
Nanopore Sequencing	11.29	13.20	15.42	18.01	21.05	24.67	29.00	34.21	17.3%
Other Technologies	80.97	91.63	103.64	117.32	132.92	151.11	172.40	197.53	13.8%
Total	220.03	250.17	284.31	323.39	368.16	420.61	482.21	555.24	14.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.2 SEQUENCING BY SYNTHESIS (SBS)

8.2.1 SBS TECHNOLOGY ENABLES HIGHEST PRODUCTION OF BASE PAIRS IN SHORT TIME

The SBS technology uses four fluorescently labeled nucleotides to sequence millions of clusters on the flow cell surface in parallel. During every sequencing cycle, a single-labeled deoxyribonucleoside triphosphate (dNTP) is added to the nucleic acid chain. This nucleotide label serves as a terminator for polymerization. After dNTP incorporation, the dye is first imaged to identify the base, and then it is enzymatically cleaved to allow the incorporation of another nucleotide. Since all four reversible terminator-bound dNTPs are present during each sequencing cycle, natural competition minimizes incorporation bias. The result is true base-by-base sequencing that enables the most accurate data for a broad range of applications.

The SBS technology has a wide customer base owing to its well-established precision in sequencing as well as its broad application base. It has applications in microbial genetics and metagenomics and is mainly used in whole-genome sequencing, transcriptome analysis, targeted sequencing, and genome-wide protein-nucleic acid interaction analysis.

9.1 INTRODUCTION

Based on end user, the global microbiome sequencing services market is segmented into pharmaceutical and biotechnology companies, academic and research institutes, and other end users (including small-scale CROs, CDMOs, hospitals, and clinics). The academic and research institutes segment accounted for the largest market share of 45.0% in 2022. This can be attributed to the high adoption of microbiome sequencing services and clinical research in these facilities.

TABLE 71 MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	70.17	80.20	91.62	104.76	119.89	137.69	158.69	183.68	14.9%
Academic and Research Institutes	99.25	112.66	127.82	145.15	164.97	188.15	215.34	247.53	14.1%
Other End Users	50.62	57.31	64.87	73.48	83.30	94.76	108.18	124.03	13.8%
Total	220.03	250.17	284.31	323.39	368.16	420.61	482.21	555.24	14.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

9.2 PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES

9.2.1 ROBUST FUNDING FOR MICROBIOME-BASED THERAPEUTICS TO DRIVE MARKET

This segment is inclusive of the application of microbiome sequencing services in drug discovery, development, and clinical diagnostics. The microbiota plays an important role in the efficacy of therapeutic compounds in drug discovery. Many therapeutic compounds are processed by the catalytic functions of enzymes encoded by mammalian symbiotic bacteria. The unique nature of the human gut microbiome offers a new source of biomarkers to distinguish between patients' physiology and diseases for enhancement in personalized medicine.

Several diseases and conditions, such as diabetes, obesity, cancer, and autoimmune disorders, have been proposed to be treatable with the help of microbiome-based products. Several animal studies, preliminary human studies, uncontrolled studies, and anecdotal observations have shown positive results for these products. For instance, human trials indicating *Lactobacillus* GG may be useful in alleviating joint pains among rheumatoid arthritis (an autoimmune disease) patients. In addition, there are several animal studies that show that probiotics inhibit the initiation or progression of colon and bladder cancers. Furthermore, in vitro, cell culture, and animal studies have indicated that probiotics bind and prevent the absorption of aflatoxins, which have been associated with the occurrence of liver cancer in humans. This has created a significant demand for microbiome sequencing services among pharmaceutical and biotechnology companies.

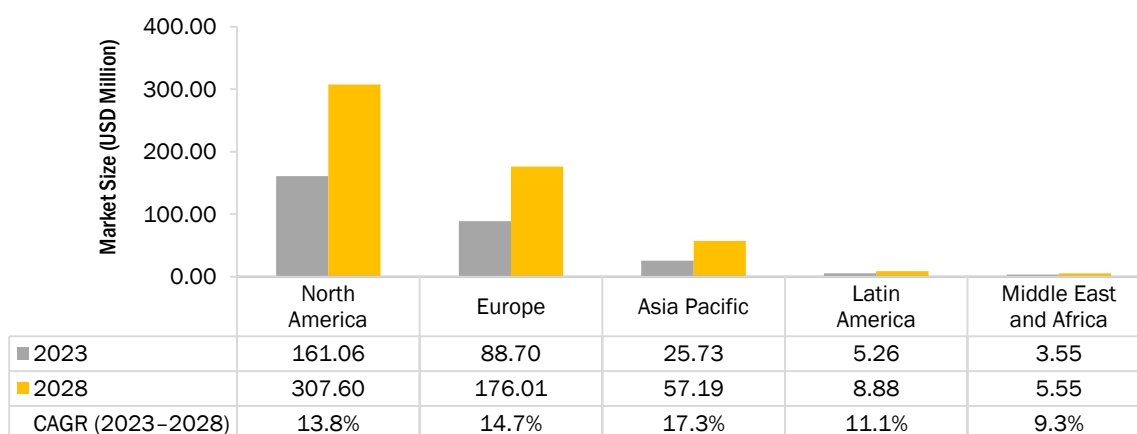
The availability of funding for research is a key factor driving market growth, considering the rising pace of activity in this field. In the last five years, the NIH has extended funding of around USD 58.3 million to boost research in the human microbiome area, including funding for developing vaccines and devices. Of this, USD 49.0 million was marked for developing therapeutic products like prebiotics, probiotics, and dietary supplements and microbiome-based novel antimicrobials (Source: NIH).

10.1 INTRODUCTION

The global microbiome sequencing services market has been segmented into five major regional segments, namely, North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. In 2022, North America dominated the market with the largest share of 56.9%, followed by Europe (31.1%). Factors such as the established research infrastructure, technological advancements, demand for precision medicine, supportive regulatory environment, robust biotechnology and healthcare industry, and availability of funding and investment are driving the growth of the microbiome sequencing services market in North America.

The microbiome sequencing services market in the Asia Pacific is estimated to register the highest growth rate of 17.3% during the forecast period, primarily due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine.

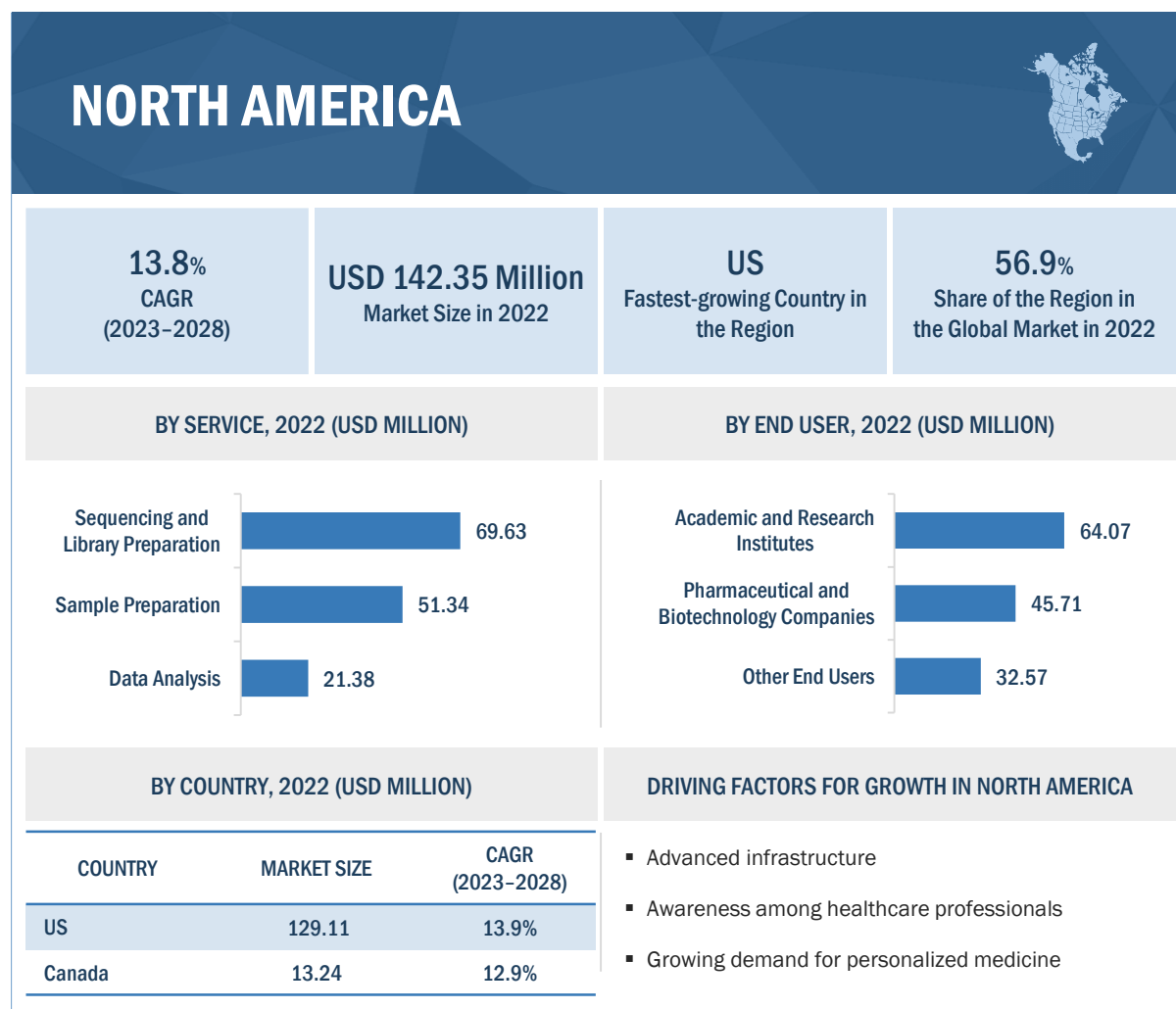
FIGURE 25 MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

North America is home to several key companies that play a pivotal role in the microbiome sequencing services market. These companies, including Charles River Laboratories, CosmosID, and CD Genomics, are involved in the development and commercialization of microbiome sequencing technologies, platforms, and services. They are significantly contributing to the growth and innovation in the North American microbiome sequencing services market through their research, product offerings, and collaborations with academic and industry partners.

FIGURE 26 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET SNAPSHOT



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Centre of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 104 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	30.83	35.22	40.20	45.94	52.54	60.30	69.44	80.32	14.8%
Sequencing by Ligation	8.75	9.96	11.32	12.89	14.69	16.80	19.27	22.21	14.4%
Nanopore Sequencing	3.49	4.12	4.86	5.72	6.74	7.96	9.43	11.21	18.2%
Other Technologies	25.14	28.51	32.32	36.67	41.63	47.44	54.24	62.28	14.0%
Total	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 105 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	21.68	24.91	28.62	32.90	37.85	43.70	50.62	58.90	15.5%
Academic and Research Institutes	30.81	35.05	39.86	45.38	51.70	59.10	67.81	78.13	14.4%
Other End Users	15.73	17.84	20.22	22.94	26.05	29.69	33.94	38.98	14.0%
Total	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

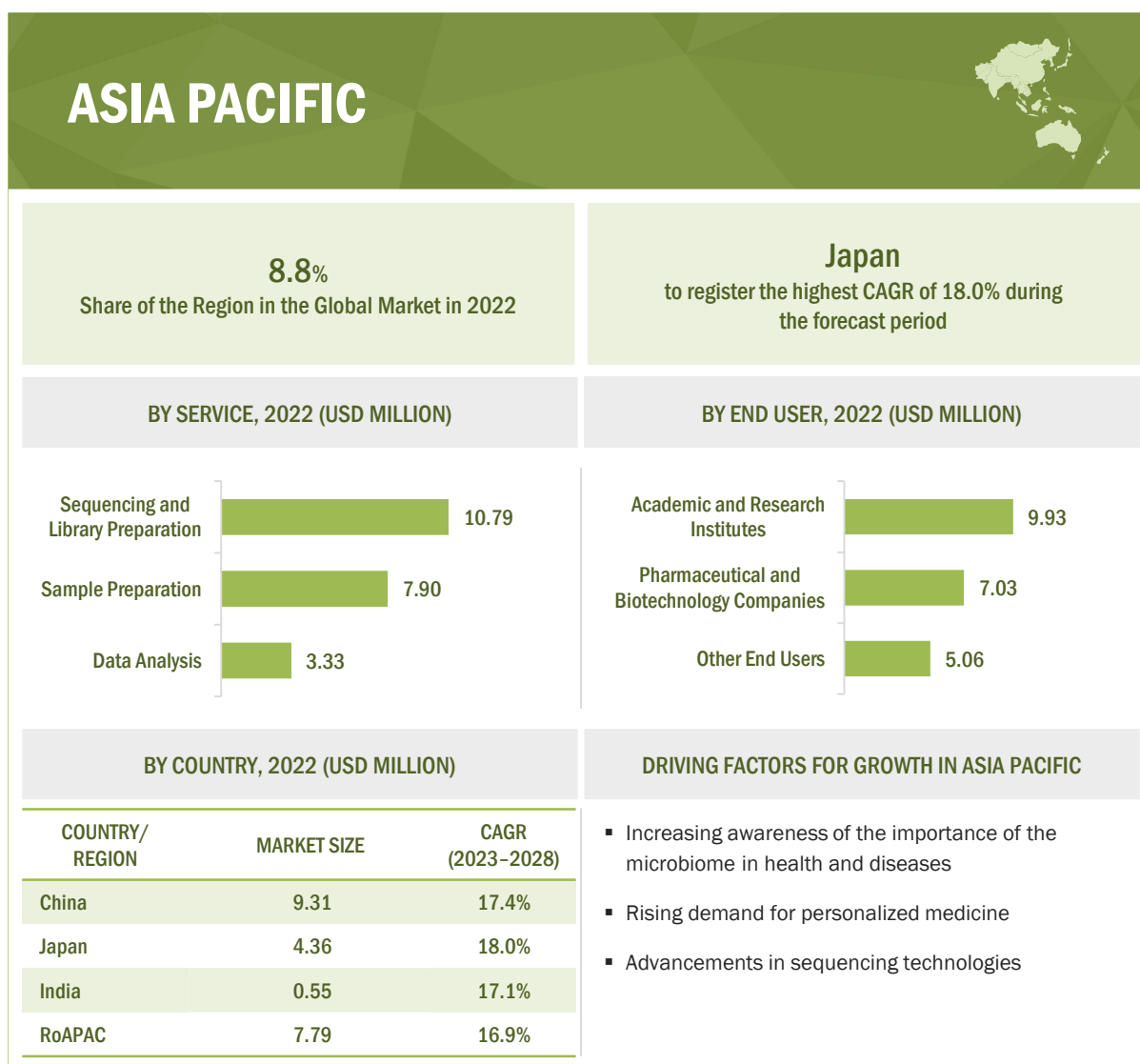
10.3.1 GERMANY

10.3.1.1 Rapidly increasing government funding to support growth

Germany is the largest contributor to the European microbiome sequencing services market, with an estimated share of 21.8% in 2023. The microbiome sequencing services market in Germany is estimated to grow from USD 19.31 million in 2023 to USD 38.88 million by 2028, at a CAGR of 15.0%.

The microbiome sequencing services market in Germany is growing rapidly, driven by the increasing demand for microbiome analysis in research and clinical applications. The German government is making significant investments in the development of innovative technologies and research infrastructure, including microbiome research. This funding is driving the growth of the microbiome sequencing services market, as it allows researchers and companies to access the resources they need to advance in the field. Also, advances in sequencing technologies and data analysis tools are making microbiome analysis faster, more accurate, and more accessible.

The high cost of sequencing technologies and the lack of standardization in data analysis and interpretation are expected to restrain the growth of this market. Despite these challenges, there are significant opportunities for growth in the German microbiome sequencing services market, including the

FIGURE 27 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET SNAPSHOT

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

11 COMPETITIVE LANDSCAPE

11.1 OVERVIEW

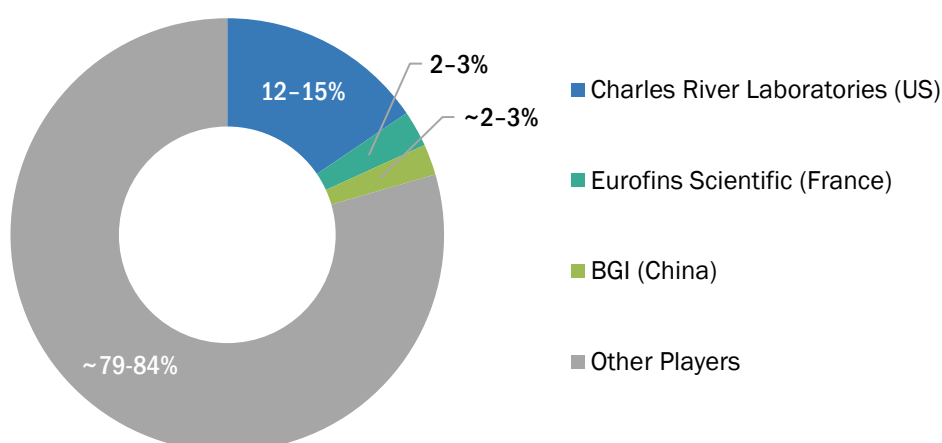
The competitive landscape includes an analysis of the key growth strategies adopted by major players between October 2020 and April 2023. Players in the microbiome sequencing services market have employed various strategies to expand their footprints and increase their market shares. The key strategies adopted by major players in the microbiome sequencing services market include expansions, service launches, acquisitions, agreements, and collaborations.

Prominent players in this market include Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

11.3 MARKET SHARE ANALYSIS

The global microbiome sequencing services market is a fragmented market, with Charles River Laboratories (US), Eurofins Scientific (France), and BGI (China) together accounting for ~20% of the global market. Players such as CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and MacroGen, Inc. (South Korea) accounted for the remaining 75–80% of the total market.

FIGURE 29 MICROBIOME SEQUENCING SERVICES MARKET SHARE ANALYSIS, BY KEY PLAYER, 2022



Other players include CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and MacroGen, Inc. (South Korea).

Source: Annual Reports, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 169 MICROBIOME SEQUENCING SERVICES MARKET: DEGREE OF COMPETITION

DEGREE OF COMPETITION	FRAGMENTED
Total Market Share of the Top 3 Players	~20%
Other Players	~80%
THE DEGREE OF COMPETITION IS DEFINED AS BELOW:	
Fragmented: When the top 5 players have a total market share of <25%	
Competitive: When the top 5 players have a total market share of 25–50%	
Consolidated: When the top 5 players have a total market share of >50%	

Note: Market shares of the above companies are calculated based on relevant segmental revenues generated by the companies.
Source: Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets

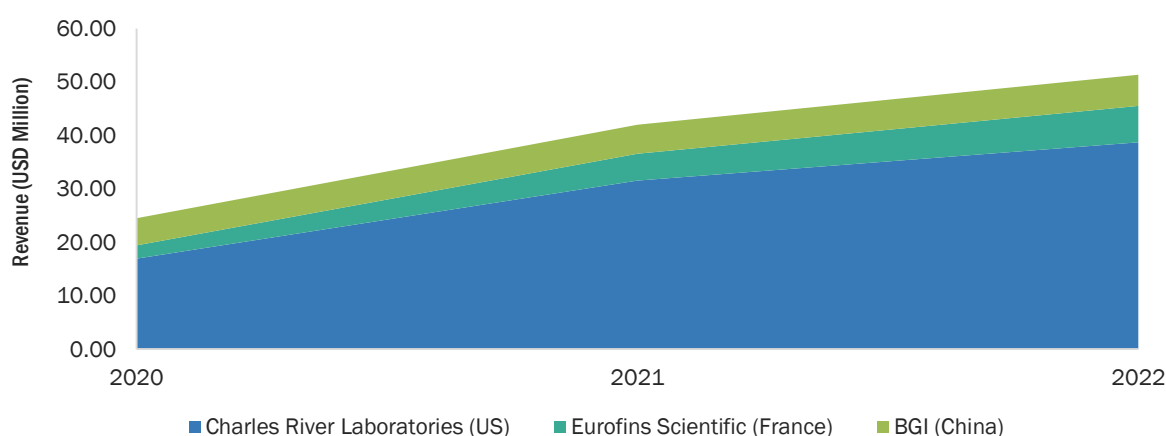
Charles River Laboratories (US) is the leading player in the microbiome sequencing services market. CRL provides superior, flexible, and tailored solutions through its comprehensive portfolio of drug discovery and research models and services. The company has been focusing on organic and inorganic growth through expansions and partnerships. For instance, CRL partnered with Fios Genomics Ltd. (UK) to gain access to its bioinformatics solutions, helping CRL to develop a full spectrum of microbiome profiling/analysis services. In addition, the strong reach of CRL's microbial detection product, Accugenix, is contributing to the company's dominant position in the market.

Eurofins Scientific (France) is a leading player in the global microbiome sequencing services market. The company's strong focus on a broad range of customers, technology-enabled solutions, investments in joint ventures and partnerships, and comprehensive microbiome sequencing solutions and processes are factors contributing to the growth of the company's market share. The company provides drug discovery and development services to more than 220,000 customers, including managed care organizations (MCOs), biopharmaceutical companies, governmental agencies, physicians, hospitals and health systems, food & nutritional companies, and independent clinical laboratories.

11.4 REVENUE SHARE ANALYSIS OF TOP MARKET PLAYERS

The revenue analysis was calculated based on the performance of major players during the past few years and the current years in the microbiome sequencing services market. This gave an estimation of the overall market size, structure, and recent advancements in service technologies. Leading players, such as Charles River Laboratories (US), Eurofins Scientific (France), and BGI (China), influence the microbiome sequencing services market due to their vast services portfolios, recent acquisitions, and wide distribution & sales networks across the globe.

FIGURE 30 REVENUE SHARE ANALYSIS OF TOP PLAYERS IN MICROBIOME SEQUENCING SERVICES MARKET, 2020-2022



Notes: The segmental revenue for the top three players have been considered. The segmental share of microbiome sequencing services may vary significantly from company to company, depending on individual company portfolios.

Source: Press Releases and Company Websites

11.5 COMPANY EVALUATION QUADRANT

The company evaluation quadrant provides information on the top 20 players offering microbiome sequencing services and outlines findings on the performance of each vendor. The industry matrix evaluations are based on two broad categories—Service Offerings and Business Strategies. Each type carries various criteria based on which the companies have been evaluated. The evaluation criteria considered under service offerings include the breadth of offerings, services launched, service technologies, and support. The evaluation criteria considered under the business strategy includes reach (based on market and geographic presence), industry coverage (based on end users catered to), viability, and inorganic growth strategies. The top 20 companies are placed into four categories based on their performance in each criterion—“Stars,” “Emerging Leaders,” “Pervasive Players,” and “Participants.”

We have evaluated 20 companies in this section, which include Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and MacroGen, Inc. (South Korea).

11.5.1 STARS

Players who fall into this category receive high scores for most evaluation criteria. They have a strong and established product portfolio, high market shares, comprehensive application coverage, and a strong geographical footprint. These players provide mature, innovative, and reputable microbiome sequencing services. Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), and QIAGEN (Germany) are the star players in this market.

11.5.2 EMERGING LEADERS

Emerging leaders have demonstrated substantial product innovations compared to their competitors (resulting in an evolved service portfolio) and significant market presence. However, they do not have robust growth strategies for geographical penetration and overall business growth. Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), and OraSure Technologies (US) are emerging leaders in the market.

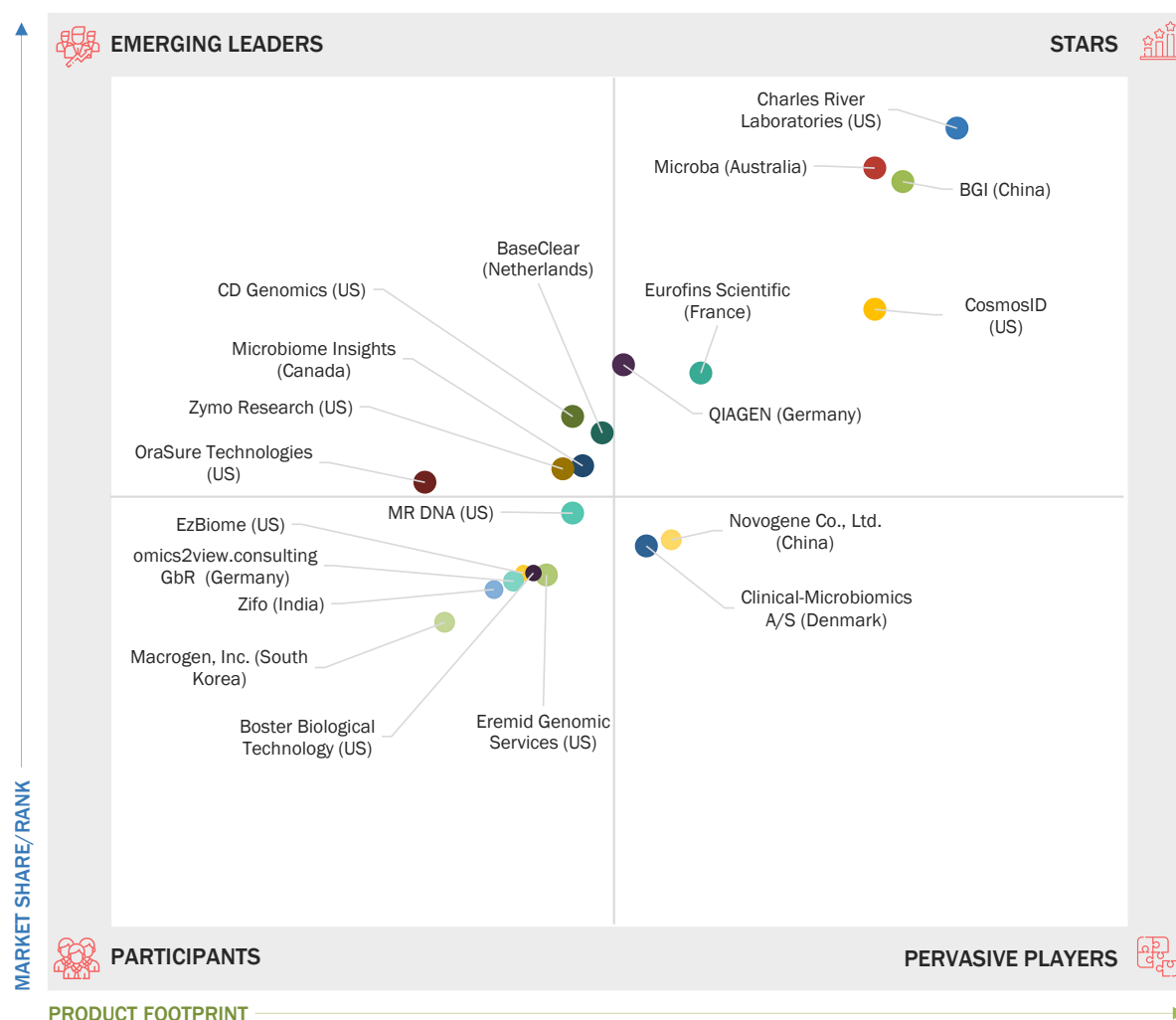
11.5.3 PERVASIVE PLAYERS

Companies falling in this quadrant are established players with very strong business strategies and global distribution/product availability. However, they have a weaker service portfolio and lower market share than the emerging leaders. Pervasive players are Clinical-Microbiomics A/S (Denmark) and Novogene Co., Ltd. (China).

11.5.4 PARTICIPANTS

The companies that fall in this category have niche service offerings. They do not have substantial business strategies as compared to other established vendors. These companies could be new entrants and require more time before getting traction. Participants in the microbiome sequencing services market include MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and MacroGen, Inc. (South Korea).

FIGURE 31 MICROBIOME SEQUENCING SERVICES MARKET: COMPANY EVALUATION QUADRANT, 2022



Source: MarketsandMarkets Analysis

13 APPENDIX

13.1 DISCUSSION GUIDE

- Q. 1.** What are your views on the growth prospects of the microbiome sequencing services market? What is the current scenario, and at what rate is the market expected to grow in the future?

Primary's Viewpoint: _____

- Q. 2.** What are your views on the major drivers, restraints, opportunities, and challenges for the microbiome sequencing services market (global/country level)?

Primary's Viewpoint: _____

- Q. 3.** What is the market size, in terms of value, of the major microbiome sequencing services in 2022 (USD million), and at what rate is the market expected to grow in the coming five years (2023–2028)?

Primary's Viewpoint: _____

- Q. 4.** Kindly provide the market shares of the following key players in the microbiome sequencing services market. Kindly validate the key players and let us know if any major players are missing.

COMPANY NAME	MARKET SHARE (2022)
Charles River Laboratories (US)	
Eurofins Scientific. (France)	
BGI (China)	
Other Players	

- Q. 5.** Kindly provide the market shares and projected growth rates of the following service segments in the microbiome sequencing services market. What are the key factors that will drive the fastest-growing market segment in the next five years? Kindly validate the market segmentation and let us know if any segments are missing or overlapping.

SERVICE	MARKET SHARE (2022)	CAGR (2023–2028)
Sample Preparation		
Sequencing and Library Preparation		
Data Analysis		

Q. 6. Kindly provide the market shares and projected growth rates of the following type segments in the microbiome sequencing services market. What are the key factors that will drive the fastest-growing market segment in the next five years? Kindly validate the market segmentation and let us know if any segments are missing or overlapping.

TYPE	MARKET SHARE (2022)	CAGR (2023–2028)
Amplicon Sequencing		
Whole-genome Sequencing		
Shotgun Sequencing		
Other Types (De Novo Sequencing and Metagenomic/Metatranscriptomic Sequencing)		

Q. 7. Kindly provide the market shares and projected growth rates of the following technology segments in the microbiome sequencing services market. What are the key factors that will drive the fastest-growing market segment in the next five years? Kindly validate the market segmentation and let us know if any segments are missing or overlapping.

TECHNOLOGY	MARKET SHARE (2022)	CAGR (2023–2028)
Sequencing by Synthesis		
Sequencing by Ligation		
Nanopore Sequencing		
Other Technologies (Semiconductor Sequencing and Sanger Sequencing)		

Q. 8. Kindly provide the market shares and projected growth rates of the following end-user segments in the microbiome sequencing services market. What are the key factors that will drive the fastest-growing market segments in the next five years? Kindly validate the market segmentation and let us know if any segments are missing or overlapping.

END USER	MARKET SHARE (2022)	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies		
Academic and Research Institutes		
Other End Users (Hospitals, Clinics, CROs, and CDMOs)		

Q. 9. Kindly provide the market shares and projected growth rates of the following regions in the microbiome sequencing services market.

REGION	MARKET SHARE (2022)	CAGR (2023–2028)
North America		
Europe		
Asia Pacific		
Latin America		
Middle East and Africa		

Q. 10. Kindly provide the market shares and projected growth rates of the following countries in the microbiome sequencing services market.

NORTH AMERICA	MARKET SHARE (2022)	CAGR (2023–2028)
US		
Canada		

EUROPE	MARKET SHARE (2022)	CAGR (2023–2028)
Germany		
UK		
France		
Italy		
Spain		
Rest of Europe		

ASIA PACIFIC	MARKET SHARE (2022)	CAGR (2023–2028)
China		
India		
Japan		
Rest of Asia Pacific		

LATIN AMERICA	MARKET SHARE (2022)	CAGR (2023–2028)
Brazil		
Rest of Latin America		

Q. 11. Kindly provide additional comments/suggestions, if any.

Primaries Viewpoint: _____